

Data Cleaning & Reporting Automation

This project demonstrates data cleaning and automated reporting techniques using Python and Pandas. The workflow includes handling missing values, removing duplicates, generating reports, and visualizing processed data.

Objectives

- Clean inconsistent and incomplete datasets
- Remove duplicate records
- Automate reporting workflows
- Generate visual summaries

Sample Raw Dataset

Employee	Department	Salary	Performance
John	HR	40000.0	78.0
Alice	IT	50000.0	90.0
Bob	Finance	NULL	85.0
Alice	IT	50000.0	90.0
NULL	HR	42000.0	NULL

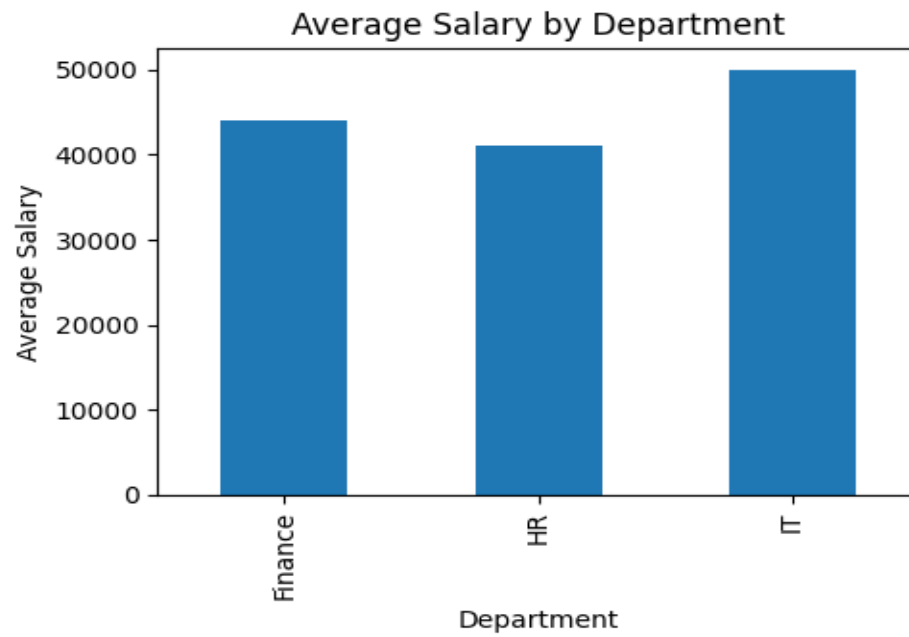
Data Cleaning Steps

1. Removed duplicate rows.
2. Filled missing employee names with 'Unknown'.
3. Replaced missing salary values with average salary.
4. Replaced missing performance values with average performance score.
5. Generated automated summary reports and charts.

Cleaned Dataset

Employee	Department	Salary	Performance
John	HR	40000.0	78.0
Alice	IT	50000.0	90.0
Bob	Finance	44000.0	85.0
Unknown	HR	42000.0	84.33

Visualization



Conclusion

The project successfully automated data cleaning and reporting tasks. It improved data quality by handling missing and duplicate values while generating useful reports and charts automatically.